

PRINCESS JULIANA INTL, Sint Maarten (Dutch part)

WMO: 788660

Lat: 18.046N

Lon: 63.115W

Elev: 14

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00399

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	71.5	73.0	60.4	78.5	76.0	62.2	83.7	76.4	22.3	78.3	20.7	78.3	7.7	90	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	89.9	79.5	89.4	79.1	88.2	78.5	81.8	87.7	80.8	86.8	80.1	86.2	10.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.3	157.5	86.5	79.0	150.8	85.8	78.5	147.9	85.4	45.5	88.0	44.6	87.0	43.7	86.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	18.3	16.7	DB	69.5	92.0	1.9	3.0	68.2	94.2	67.1	96.0	66.1	97.7	64.7	99.9	(4)
(5)			WB	63.1	84.2	1.9	1.9	61.8	85.6	60.7	86.6	59.7	87.7	58.3	89.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	78.6	78.4	79.0	80.2	81.8	83.3	83.7	84.2	84.2	83.3	81.6	79.9	(6)	
	DBStd	2.63	1.40	1.50	1.53	1.54	1.61	1.32	1.48	1.43	1.55	1.68	1.76	1.66	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11507	886	796	899	906	985	1000	1046	1060	1025	1032	947	925	(10)	
	CDD65	6032	421	376	434	456	520	550	581	595	575	567	497	460	(11)	
	CDH74	62339	3126	2706	3369	4168	5494	6393	6948	7274	7034	6635	5167	4025	(12)	
	CDH80	18407	383	327	520	859	1482	2190	2569	2869	2790	2348	1374	696	(13)	
(14) Wind	WSAvg	10.2	10.7	10.9	10.3	10.0	10.0	10.9	11.7	10.6	8.8	8.5	9.1	10.6	(14)	
(15) Precipitation	PrecAvg	41.6	2.5	1.9	1.8	2.5	3.9	2.4	3.2	4.0	4.8	5.0	5.5	3.9	(15)	
	PrecMax	70.9	6.7	6.6	6.7	10.8	13.6	7.5	8.3	9.2	14.3	14.2	23.6	12.0	(16)	
	PrecMin	27.7	0.1	0.3	0.3	0.5	0.3	0.1	0.3	1.0	1.0	1.2	1.0	0.6	(17)	
	PrecStd	10.0	1.3	1.2	1.2	1.9	3.1	1.7	1.7	1.8	2.7	3.0	3.9	2.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.6	84.6	85.8	87.2	88.6	89.8	90.0	91.0	91.3	90.0	88.2	86.8	(19)	
		MCWB	75.6	74.9	74.4	77.0	78.8	79.8	79.6	80.6	81.2	80.0	78.3	78.1	(20)	
	2%	DB	84.1	83.9	84.4	86.1	87.7	88.2	89.4	89.7	89.8	89.1	87.1	85.7	(21)	
		MCWB	75.0	74.3	74.0	76.2	78.4	78.5	79.0	79.2	79.4	79.3	77.6	76.7	(22)	
	5%	DB	82.8	82.7	83.9	84.9	86.4	87.8	88.1	88.7	89.2	88.0	86.2	84.4	(23)	
		MCWB	73.7	73.3	73.8	75.4	77.1	78.3	78.1	78.7	79.0	78.6	77.0	75.6	(24)	
	10%	DB	82.3	82.3	82.7	84.2	85.8	86.7	87.6	87.9	88.0	87.4	85.6	83.7	(25)	
		MCWB	73.3	73.0	73.1	74.9	76.7	77.6	77.8	78.4	78.6	78.4	76.8	75.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.5	78.5	77.8	79.2	80.7	81.7	82.2	82.8	83.8	82.8	80.5	80.3	(27)	
		MCDB	83.5	83.0	82.9	85.1	86.7	88.3	88.6	88.8	89.1	88.0	85.1	84.6	(28)	
	2%	WB	76.5	76.7	76.2	77.9	79.6	80.3	80.8	81.0	81.1	81.0	79.5	78.2	(29)	
		MCDB	82.2	81.9	82.2	84.0	85.7	86.9	86.9	87.0	87.6	86.5	84.7	83.3	(30)	
	5%	WB	75.4	75.2	75.2	77.0	78.6	79.5	79.8	80.1	80.1	80.0	78.6	77.0	(31)	
		MCDB	81.5	81.1	81.8	83.2	84.7	85.9	86.0	86.2	86.7	85.8	84.3	82.7	(32)	
	10%	WB	74.3	74.0	74.3	76.0	77.8	78.7	79.0	79.4	79.4	79.2	77.7	76.0	(33)	
		MCDB	80.5	80.3	81.0	82.5	83.9	85.0	85.3	85.7	86.0	85.2	83.6	81.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.2	8.4	8.4	7.9	7.8	8.1	8.2	8.3	8.3	8.3	8.0	(35)	
		MCDBR	8.7	8.4	9.0	9.0	8.5	8.6	8.7	8.8	9.4	8.9	8.6	8.5	(36)	
	5% WB	MCWBR	5.0	4.8	4.8	4.8	4.5	4.5	4.3	4.4	4.8	4.7	4.5	4.7	(37)	
		MCWBR	8.5	8.2	7.9	8.2	8.0	8.4	8.4	8.3	8.3	8.1	7.9	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.360	0.368	0.371	0.398	0.439	0.506	0.512	0.475	0.433	0.387	0.371	0.366	(40)	
		taud	2.613	2.585	2.573	2.518	2.421	2.187	2.163	2.308	2.458	2.603	2.644	2.627	(41)	
	Ebn at Noon	Ebn at Noon	294	297	298	288	273	253	252	263	275	287	289	289	(42)	
		Edh at Noon	29	31	32	35	38	47	48	42	36	30	28	28	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1579	1815	2030	2131	2064	2092	2100	2084	1938	1720	1520	1460	(44)	
	RadStd	RadStd	36	34	67	101	118	106	105	71	100	109	90	61	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page